Customer Shopping Behavior Analysis

Project Type: Data Analytics

Tools Used: Python, SQL, Power BI

Dataset: Customer Shopping Behavior Dataset

1.Project Overview

This project analyzes customer shopping behavior using Python, SQL, and Power BI. The objective is to understand customer spending patterns, product preferences, discount usage, subscription behavior, and purchasing frequency. The analysis aims to identify meaningful business insights and provide data-driven recommendations to improve customer engagement and business performance.

2. Dataset Summary

Dataset Details :

- **Total Records:** 3,900
- **Total Columns:** 18
- **Missing Values:** 37 in the Review Rating column.

Key Features:

- **Customer Demographics:** Age, Gender, Location, Subscription Status.
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color.
- **Shopping Behavior:** Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type.

3. Exploratory Data Analysis Using Python

Python was used to load, explore, and clean the customer shopping behavior dataset. The Pandas library helped examine the dataset structure, identify missing values, standardize column names, and create new features for further analysis.

3.1 Data Loading and Initial Exploration

- Loaded the dataset into Python using the Pandas library.
- Used df.head() to preview the first five rows.
- Used df.info() to examine column names, data types, and non-null values.
- Used df.describe() to understand numerical columns through summary statistics.

3.2 Missing Data Handling

The dataset was checked for missing values using Pandas. The Review Rating column contained 37 missing values. These missing values were filled using the median review rating of the corresponding product category. This approach helped retain the records while handling missing ratings.

```
[38]: df.isnull().sum()
```

```
[38]: Customer ID                0
      Age                        0
      Gender                     0
      Item Purchased             0
      Category                   0
      Purchase Amount (USD)      0
      Location                   0
      Size                       0
      Color                      0
      Season                     0
      Review Rating             37
      Subscription Status        0
      Shipping Type              0
      Discount Applied           0
      Promo Code Used            0
      Previous Purchases         0
      Payment Method             0
      Frequency of Purchases     0
      dtype: int64
```

```
[39]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x : x.fillna(x.median()))
```

```
[40]: df.isnull().sum()
```

```
[40]: Customer ID                0
      Age                        0
      Gender                     0
      Item Purchased             0
      Category                   0
      Purchase Amount (USD)      0
      Location                   0
      Size                       0
      Color                      0
      Season                     0
      Review Rating              0
      Subscription Status        0
      Shipping Type              0
      Discount Applied           0
      Promo Code Used            0
      Previous Purchases         0
      Payment Method             0
      Frequency of Purchases     0
      dtype: int64
```

3.3 Column Standardization

The column names were converted to lowercase, and spaces were replaced with underscores to maintain consistency and improve readability during analysis.

```
[41]: df.columns = df.columns.str.lower()
      df.columns = df.columns.str.replace(' ','_')
      df = df.rename(columns = {'purchase_amount_(usd)':'purchase_amount'})
```

```
[42]: df.columns
```

```
[42]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
             'purchase_amount', 'location', 'size', 'color', 'season',
             'review_rating', 'subscription_status', 'shipping_type',
             'discount_applied', 'promo_code_used', 'previous_purchases',
             'payment_method', 'frequency_of_purchases'],
            dtype='str')
```

3.4 Feature Engineering

Two new features, age_group and purchase_frequency_days, were created to support customer segmentation and purchasing behavior analysis.

```
[43]: # create a column age_group
      labels = ['Young Adult','Adult','Middle-aged','Senior']
      df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
[44]: df[['age','age_group']].head(10)
```

[44]:

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

```
[45]: # create a column purchased_frequency_days
      frequency_mapping = {
          'Fortnightly': 14,
          'Weekly': 7,
          'Monthly': 30,
          'Quarterly': 90,
          'Bi-Weekly': 14,
          'Annually': 365,
          'Every 3 Months': 90
      }

      df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
[46]: df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

[46]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

3.5 Data Consistency Check

The discount_applied and promo_code_used columns were compared for consistency. As both columns contained identical values in the dataset, promo_code_used was removed as a redundant column.

```
[47]: df[['discount_applied','promo_code_used']].head(10)
```

[47]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
[48]: (df['discount_applied'] == df['promo_code_used']).all()
```

```
[48]: np.True_
```

```
[49]: # Dropping promo code used column

df = df.drop('promo_code_used', axis=1)
```

```
[50]: df.columns
```

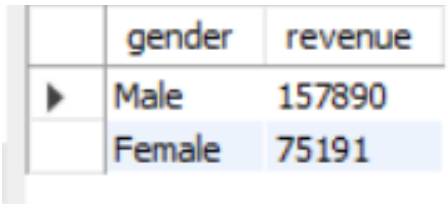
```
[50]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

4. SQL Analysis

SQL was used to analyze customer shopping behavior and answer business-related questions. The analysis focused on revenue, customer spending, product ratings, shipping preferences, subscription status, discount usage, customer segmentation, and purchasing patterns.

4.1 Revenue by Gender

This query calculates the total purchase revenue generated by male and female customers to compare their spending contribution.



gender	revenue
Male	157890
Female	75191

4.2 High-Spending Discount Users

Identify customers who used discounts but spent more than the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94

4.3 Top 5 Products by Average Review Rating

Identified the five products with the highest average review ratings.

item_purchased	Average_Product_Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4.4 Average Purchase Amount by Shipping Type

Compare average purchase amounts for Standard and Express shipping.

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

4.5 Subscribers vs. Non-Subscribers

Compare customer count, average spending, and total revenue by subscription status.

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

4.6 Products with the Highest Discount Rate

Find the five products with the highest percentage of records where a discount was applied.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4.7 Customer Segmentation

Classify customers into New, Returning, and Loyal segments based on previous purchases.

customer_segment	Number_of_Customers
Loyal	3116
Returning	701
New	83

4.8 Top 3 Products in Each Category

Identify the three most frequently appearing products in each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4.9 Repeat Buyers and Subscription Status

Examine subscription status among customers with more than five previous purchases.

subscription_status	repeat_buyers
Yes	958
No	2518

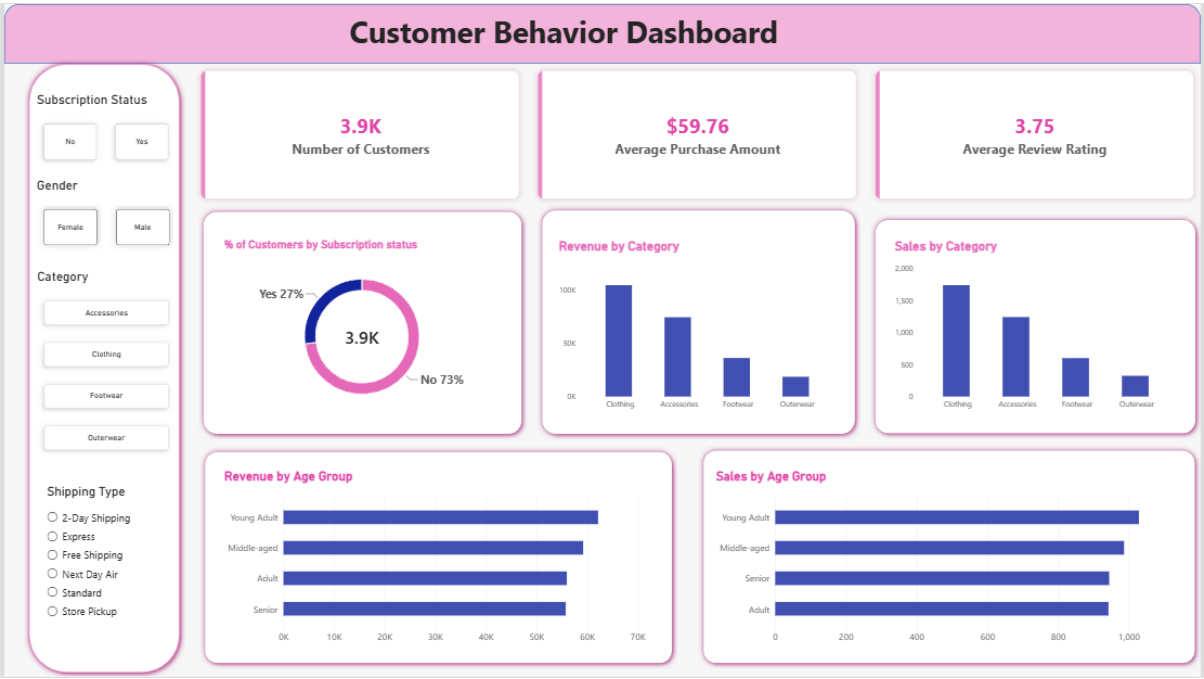
4.10 Revenue by Age Group

Calculate total purchase revenue for each age group.

	age_group	total_revenue
▶	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Power BI Dashboard

An interactive dashboard was developed using Power BI to visualize customer shopping behavior. It presents key performance indicators and charts related to customer spending, subscription status, product categories, age groups, and shopping preferences. Slicers allow users to explore the data using different filters.



6. Key Insights

- **Revenue by Gender:** Male customers generated higher total purchase revenue than female customers.
- **Product Ratings:** Gloves had the highest average review rating among the top five products, at 3.86.
- **Shipping Preferences:** Express shipping had a higher average purchase amount ($60.48) than Standard shipping ($58.46).

- **Subscription Behavior:** Non-subscribers generated higher total revenue ($170,436) than subscribers ($62,645).
- **Customer Segmentation:** The analysis classified 3,116 records as Loyal, 701 as Returning, and 83 as New.
- **Discount Usage:** Hats had the highest discount application rate among the products analyzed, at 50.00%.

7. Business Recommendations

- **Improve Customer Loyalty:** Consider loyalty rewards for repeat customers.
- **Review Discount Strategy:** Evaluate discount usage across products before adjusting promotional offers.
- **Promote Highly Rated Products:** Highlight products with strong average review ratings.
- **Strengthen Subscription Engagement:** Use subscriber and non-subscriber analysis to identify opportunities for subscription growth.
- **Targeted Marketing:** Use customer spending patterns and age-group revenue to guide marketing campaigns.

8. Conclusion

This project analyzed customer shopping behavior using Python, SQL, and Power BI. Python supported data cleaning and feature engineering, SQL was used to answer business questions, and Power BI presented the analysis through an interactive dashboard. The project demonstrates how data analytics tools can be combined to explore customer behavior and support data-driven business decisions.